

JENKINS CI PIPELINE

Corrected AWS EC2 + Ubuntu Jenkins Setup Guide

This guide corrects the old commands in the previous PDF. Run the commands one after another. Each command below is shown as a complete command.

Important: Your old guide used Java 17 and the Jenkins 2023 signing key. For the current Jenkins release, this guide uses Java 21 and the 2026 signing key.

What you need

1. AWS account
2. Ubuntu EC2 instance
3. Internet access to the EC2 instance
4. Browser access to the EC2 Public IPv4 address

STEP 1 — Connect to your Ubuntu EC2 instance

Open AWS → EC2 → Instances → select your instance → Connect. After connecting, you will see a terminal prompt similar to: root@ip-... or ubuntu@ip-...

STEP 2 — Update package information

Run this command and wait until it finishes:

```
sudo apt update
```

STEP 3 — Install Java 21

Jenkins needs Java to run. Install Java 21:

```
sudo apt install fontconfig openjdk-21-jre -y
```

STEP 4 — Check the Java version

Run:

```
java -version
```

Expected result

You should see Java 21 in the output. If Java 17 is shown, Jenkins may not start with the current Jenkins version.

STEP 5 — Add the current Jenkins signing key

Copy and paste this as one complete command. Do not press Enter in the middle:

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee  
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

STEP 6 — Add the Jenkins LTS repository

Copy and paste this entire line:

```
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]  
https://pkg.jenkins.io/debian-stable binary/" | sudo tee  
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

STEP 7 — Update package information again

This downloads the package information from the Jenkins repository:

```
sudo apt update
```

STEP 8 — Install Jenkins

There must be a space before -y:

```
sudo apt install jenkins -y
```

STEP 9 — Start Jenkins

Run:

```
sudo systemctl start jenkins
```

STEP 10 — Enable Jenkins at boot

This makes Jenkins start automatically when the EC2 instance restarts:

```
sudo systemctl enable jenkins
```

STEP 11 — Check Jenkins status

Run:

```
sudo systemctl status jenkins --no-pager
```

Expected status

Look for: **Active: active (running)**

STEP 12 — Open port 8080 in AWS Security Group

In AWS: EC2 → Instances → select your Jenkins instance → Security → click the Security Group → Inbound rules → Edit inbound rules → Add rule.

Type: Custom TCP

Port: 8080

Source: My IP is safer for personal testing. Use a wider source only if your instructor/lab specifically requires it.

STEP 13 — Find the EC2 Public IPv4 address

Go back to your EC2 instance details and copy the Public IPv4 address.

STEP 14 — Open Jenkins in your browser

In the browser address bar, enter:

```
http://YOUR_PUBLIC_IPV4:8080
```

Example only: if your Public IPv4 address is 12.34.56.78, open `http://12.34.56.78:8080`

STEP 15 — Get the Jenkins initial password

Run this command in the Ubuntu terminal:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Copy the password and paste it into the Jenkins Unlock Jenkins page.

STEP 16 — Finish Jenkins setup

1. Paste the initial administrator password.
2. Click **Continue**.
3. Choose **Install suggested plugins**.
4. Wait for plugin installation.
5. Create your Jenkins administrator username and password.
6. Click **Save and Finish**.
7. Click **Start using Jenkins**.

Troubleshooting

Problem: Jenkins service failed to start

Check the detailed error:

```
sudo journalctl -xeu jenkins.service --no-pager -n 50
```

Check the Java version:

```
java -version
```

If your current Jenkins release requires Java 21 or newer but the terminal shows Java 17, install Java 21 and restart Jenkins.

If Jenkins does not open in the browser

1. Check: `sudo systemctl status jenkins`
2. Confirm Jenkins is active (running).
3. Confirm AWS Security Group allows TCP port 8080.
4. Use `http://PUBLIC-IP:8080`, not `https://PUBLIC-IP` unless you separately configured HTTPS.

Important typing mistakes to avoid

Correct: `curl -fsSL`

Wrong: `curl-fsSL`

Correct: `jenkins -y`

Wrong: `jenkins-y`

Always paste each complete command as a single command. Do not split a URL or command onto the next line.

Sources used for this corrected guide

Official Jenkins Linux installation documentation and Jenkins Java support policy, checked August 2026.